

4.5.4 Moments, levers and gears (physics only)

Content	Key opportunities for skills development
A force or a system of forces may cause an object to rotate. Students should be able to describe examples in which forces cause rotation. The turning effect of a force is called the moment of the force. The size of the moment is defined by the equation:	
<i>moment of a force = force $\times$ distance</i> [$M = F d$] moment of a force, M, in newton-metres, Nm force, F, in newtons, N distance, d, is the perpendicular distance from the pivot to the line of action of the force, in metres, m. If an object is balanced, the total clockwise moment about a pivot equals the total anticlockwise moment about that pivot. Students should be able to calculate the size of a force, or its distance from a pivot, acting on an object that is balanced. A simple lever and a simple gear system can both be used to transmit the rotational effects of forces. Students should be able to explain how levers and gears transmit the rotational effects of forces.	MS 3c Students should be able to recall and apply this equation.